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Power

by Vanita K

FILE	POWER-ELECTRONICS-REPORT.PDF (1.26M)		
TIME SUBMITTED	16-FEB-2017 12:01AM	WORD COUNT	6357
SUBMISSION ID	67248792	CHARACTER COUNT	32460

Imperial College
London

Power Electronics

EE3-14 Coursework Part 1

System Analysis, Simulation and Design

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Appendix I: Appendix 0

1. Introduction

In the first part of the Power Electronic coursework, we will aim to design and simulate a boost SMPS with ideal components and real components. We will first look at the design in open loop, before constructing a compensator to study the closed loop characteristics of the converter.

2. Basic Converter Design

The design parameters for the boost converter are described as follows:

TABLE 1: Design Parameters for Boost Converter

V_{in}	$20V(\pm 2)$
V_{out}	$48V(\pm 0.5)$
$I_{out,max}$	0.6A
$\Delta I_{in,max}$	10% at $P_{out,max}$
F_{max}	200kHz

Using the above values, it is possible to calculate Δ and I_{in}

$$\delta = \frac{V_{out} - V_{in}}{V_{out}} \in [0.537, 0.629]$$

$$\Delta I_{in,max} = \frac{V_{out,max} \times I_{out,max}}{10 \times V_{in,min}} \approx 0.162A$$

2.1. Inductor Sizing

The voltage across an inductor is defined as $V_{in} = L\frac{dI_{in}}{dt}$. Rearranging this equation gives

$$\begin{aligned} L &= V_{in}\frac{dt}{dI_{in}} \\ &= V_{in}\frac{t_{on}}{\Delta I_{in}} \\ &= \frac{V_{in}}{\Delta I_{in}}\delta T \\ &= \frac{V_{in}}{\Delta I_{in}}\frac{\delta}{f} \end{aligned}$$

Correct equation

$$\therefore L_{min} = \frac{V_{in,min}}{\Delta I_{in,max}}\frac{\delta_{min}}{f_{max}} \approx 323\mu H$$

2.2. Inductor Current

The inductor current can be characterised as peak and rms current.

The worst case peak current will determine the energy stored in the inductor. When selecting an inductor, it is important for this peak current to be smaller than the saturation current, else the effective permeability of the core is reduced and effective inductance will be changed.[1]

The worst case peak inductor current assuming continuous conduction is defined as

$$I_{peak} = I_{in} + \frac{1}{2}\Delta I_{in} \approx 1.70A$$

The worst case RMS current determines the energy dissipated as heat. Changes in temperature can alter the effective inductance, therefore, it is important that the inductor selected has minimal change in inductance for the given change in temperature due to the heat dissipated by the RMS current.[1]

The worst case RMS inductor current in steady state can be defined as[2]

$$I_{rms}^2 = I_{in}^2 + \frac{1}{12}(\Delta I_{in}^2)$$

$$\therefore I_{rms_{max}} \approx 1.62A$$ Good

2.3. Capacitor Sizing

The current through the capacitor can be defined as $I_{out} = C\frac{dV_{out}}{dt}$. Rearranging this equation gives

$$\begin{aligned} C &= I_{out}\frac{dt}{dV_{out}} \\ &= I_{out}\frac{t_{on}}{\Delta V_{out}} \\ &= \frac{I_{out}}{\Delta V_{out}}\delta T \\ &= \frac{I_{out}}{\Delta V_{out}}\frac{\delta}{f} \end{aligned}$$

$$\therefore C_{min} = \frac{I_{out}}{\Delta V_{out,max}}\frac{\delta_{min}}{f_{max}} \approx 1.61\mu F$$ 2

2.4. Switch and Diode Selection

When the switch is closed, all current flows through the switch and the diode needs to block V_{out} and I_{in}.[2]

Maximum Voltage blocking capability for diode $= V_{out,max} = 48.5V$
Maximum Current blocking capability for diode $= I_{in,peak} \approx 1.70A$

When the switch is open, all current flows through the diode and the switch needs to block V_{out} and I_{in}.[2]

Maximum Voltage blocking capability for switch $= V_{out,max} = 48.5V$
Maximum Current blocking capability for switch $= I_{in,peak} \approx 1.70A$

3. Simulation of Ideal Converter in Open Loop

Using the equations derived in 2 and using $V_{in} = 20V$, the ideal converter can be simulated in PSPICE. The 100% load is obtained using $\frac{V_{out}}{I_{out}} = \frac{48V}{0.6A} = 80\Omega$.

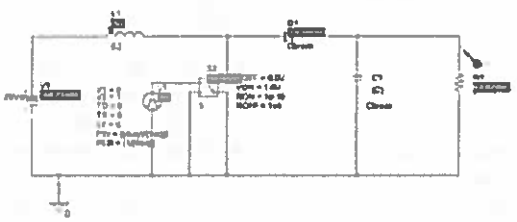


Fig. 1: Schematic of Ideal Converter in Open Loop

3.1. Efficiency with Varying Load

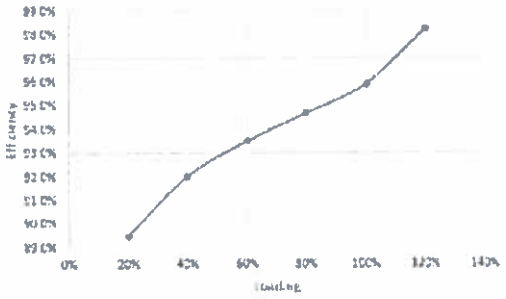


Fig. 2: Graph of Efficiency against Loading

The efficiency of the boost converter with ideal components in open loop decreases with R_{out}.

Since ideal components are used, the only power loss is due to the diode characterised as $P_{diode} = V_{DS} I_{out}$.

$$\therefore \eta \propto \frac{P_{load}}{P_{load} + P_{diode}}$$

As loading increases, current through R_{out} increases proportionally. Therefore, since $P_{load} = I^2 R$, P_{load} increases with increasing loading. Since P_{diode} is a constant for varied load, the efficiency increases for increasing loading.

3.2. Efficiency with Varying Input Voltage

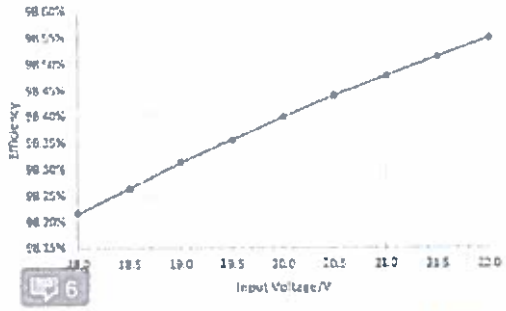


Fig. 3: Graph of Efficiency against input voltage

The efficiency of the boost converter with ideal components in open loop increases proportionally to V_{in}.

As V_{in} increases, the inductor current ripple increases as shown by the equation below, thus increasing the peak inductor current.

$$\Delta I_{in} = \frac{V_{in}\,\delta}{L\,f}$$

The peak inductor current determines the energy stored in the inductor when the switch is closed, thus an increase in the peak inductor current increases the average energy stored in the inductor, which also determines the average energy stored in the capacitor. Since ideal components are used, the only power loss is due to the diode and is constant. Therefore, an increase in input voltage would increase the output power, thus increasing efficiency.

3.3. Converter at Turn on

3.3.1 Overshoot of Output Voltage

Overshoot in a boost converter is caused by the initial inrush current required to charge the output capacitor to the input voltage.[3] The overshoot remains constant at 41.2% with change in input and output voltage but constant full load and duty cycle. However, increasing the loading (or reducing R_{out}) reduces the overshoot.

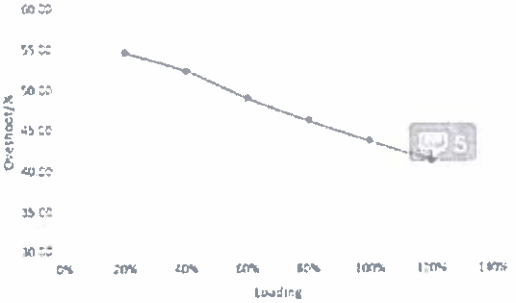


Fig. 4: Graph of Output Voltage Overshoot against Loading

This is because the magnitude of the overshoot voltage is dependent purely on the impedance of the circuit and ESR(effective series resistance) of the capacitor.[4] Since the circuit uses ideal components in this simulation, the magnitude of the overshoot relies on the line impedance. Therefore, the overshoot decreases as loading increases.

3.3.2 Rise and Settling Time of Output Voltage

The rise time is measured as time required to reach 90% of the steady state value while the settling time is measured as the time required to reach within 5% of the steady state value. As seen with the overshoot, the rise and settling times are constant for the given transfer function but varies with the loading. For the given transfer function and full load, the rise time is 0.38ms and the settling time is 3.04ms.

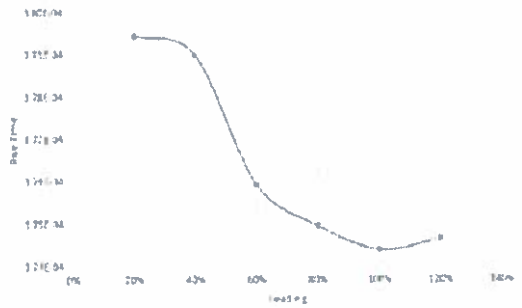


Fig. 5: Graph of Rise Time against Loading

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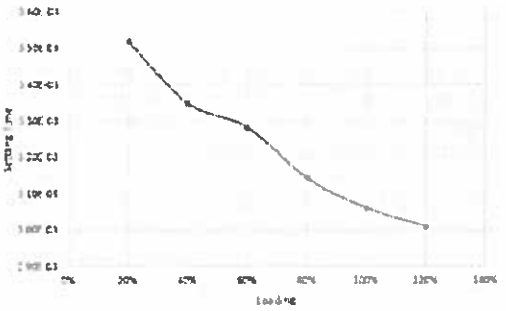


Fig. 6: Graph of Settling Time against Loading

The rise and settling time are dependent on the time constant of the RC circuit which is given by $\tau = RC$. Therefore, it is unchanged with the transfer function but decreases with increasing loading(decreasing R_{out}).

3.3.3 Oscillation Frequency of Output Voltage

The oscillation frequency of the output voltage is constant with the given transfer function but varies with loading. For full load, the oscillation frequency remains constant at 1.06kHz.

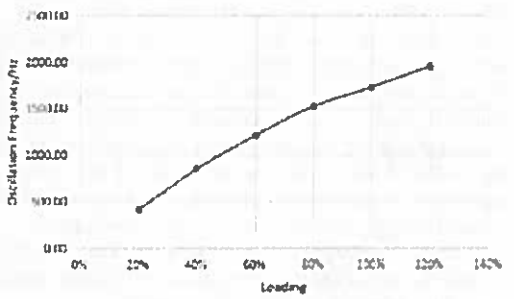


Fig. 7: Graph of Oscillation Frequency against Loading

The oscillation is caused by the resonance of the charging LC circuit, which is then discharged by the RC circuit. As seen earlier, when the loading increases, the time constant decreases, thus the oscillation frequency at the output increases, as time and frequency are inversely proportional.

3.4. Effects of Switching Loads

Figures 8 and 9 show the simulation results of switching loads between 50% and 100%. The area highlighted in blue displays the results when the load has been switched to 50% while the non-highlighted area displays the result when the load has been switched to 100%

For a fixed duty cycle, stepping of loads between 50% and 100% causes ringing that ranges up to 60V. In reality, the load will not be constant, therefore, it is not possible to provide a stable DC voltage with an open loop converter. These effects can be minimised by using a closed loop system to control the duty cycle.

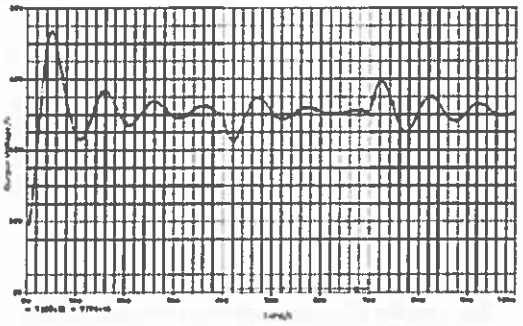


Fig. 8: Simulation of Output Volatge with Switching of Loads from 100% - 50% - 100%

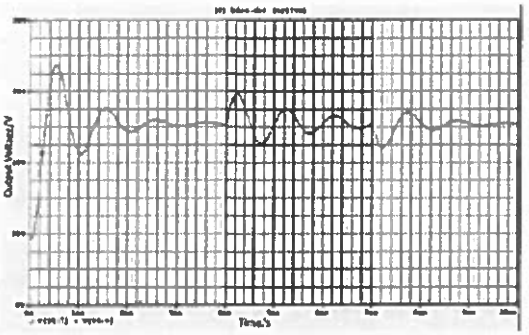


Fig. 9: Simulation of Output Volatge with Switching of Loads from 50% - 100% - 50%

3.5. Effects of Very Low Loads

At very low loads, the converter is pushed into discontinuous conduction. This is due to the fact that the high line impedance reduces the inductor current beyond $\frac{1}{2}\Delta I$, such that there are periods of time when both the switch and the diode are not conducting. The output voltage in discontinuous mode is dependent on the input voltage, duty cycle, power stage inductance, switching frequency and output load resistance,[5] therefore the output voltage will be different based on the circuit parameters.

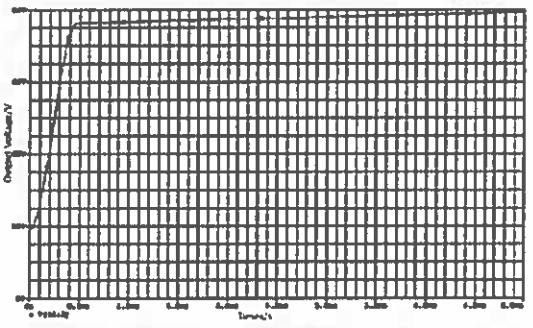


Fig. 10: Simulation of Output Volatge with Very Low Loading

4. Component Choices

In the following section, the selection of each component in the boost converter will be carried out to meet the required specification. The components are chosen from an online catalogue. In particular, the effect of each property of the components will be examined, which will form the basis of the PSPICE model of the real components.

4.1. Inductor

The selection of the inductor has proven to be particularly challenging, as the requirements on the inductor are especially stringent, as follows: (i) capable to withstand high current of about 1.7A, (ii) have a relatively large inductance of 470 μH, (iii) have maximum volume limit of $9mm^3$. For high efficiency, the losses from the inductor should be minimised, the parasitic ESR of the inductor should also be low, as well as a closed-loop iron core to minimise the leakage.

Most inductors with the suitable electrical characteristics failed to meet the volume requirement, with the exception of a *Murata Power Inductor - 60B474C*, which met the electrical and physical requirements, and also had the desired low ESR of 100 $m\Omega$.

Also, from the datasheet, the inductor has a dimension of $20.8 \times 20 \times 19.5$ mm, which gives $8112mm^3$ volume out of the $9000mm^3$ limit. The inductor also has excellent thermal properties and frequency response, which meant that the effective inductance was still around its manufactured value at our operating current and switching frequency.[6]

The low ESR allowed a high RMS current to pass through the inductor without significant power loss. The inductor design also allowed a high peak current without a significant loss of the inductance through the saturation of core.

4.2. Capacitor

The capacitor in the boost converter plays an important part acting as an output filter to the otherwise chopped current, when the MOSFET is off. Hence, to minimise the output voltage ripple, it is important to minimise the ESR of the capacitor. In order to achieve this, two capacitors - a ceramic capacitor with low ESR but also low capacitance and an eletrolytic capacitor with higher ESR and also high capacitance, will be used in parallel. This is due to the different corner frequencies of the different capacitors. We can use the ceramic capacitor to filter the high frequency ripple and the electrolytic capacitor to filter out the low frequency ripple. The capacitance value chosen is 100μF, which is much larger than the calculated minimum capacitor value found in section 2.3. This capacitor value was chosen since it does not take into account the low frequency fluctuation as a result of varying load, input voltage and the output load current variations.

For the electrolytic capacitor, *Nichicon PCR1J101MCLIGS* of value 100 μF was found to be good for these properties, with an ESR of 24 mΩ at a rated voltage of 63V. The rated ripple for the capacitor is 3Arms, as required.[7]

For the ceramic capacitor, a general purpose ceramic capacitor of 0.1μF which has an output filter corner frequency of about 700 kHz at full load was chosen. This capacitor will be useful in controlling output voltage ripple caused by parasitic inductance in the components. The typical ESR for a ceramic capacitor is very low of about 10mΩ. Hence, the ripple effect caused by this capacitor on the output voltage is small.[8] A typical ceramic capacitor is the *0.1μF TDK FK22C0G2A104J*.

4.3. MOSFET

The main requirements of the MOSFET was the V_{DS} breakdown voltage, which must be at least $V_{out} \approx 50$V and I_{DS}, which must be at least $I_{L,max} = I_{in} + \frac{1}{2}\Delta i_L \approx 1.7$A. It is also important to consider the conduction $R_{DS(on)}$ and switching losses; these should be as small as possible to maximise the efficiency.

The type of MOSFETs considered were n-channel enhancement mode MOSFETs, as the boost converter has a low-side switch and enhancement mode would be the easiest to control via a MOSFET driver.

The final selection for the MOSFET is *International Rectifier IRF530N*. This choice meets the high $I_{DS,max} = 17\text{A} > 1.7\text{A}$, as well as the breakdown $V_{DS} = 100\text{V} > 50\text{V}$, which has a low $R_{DS(on)} = 0.09\Omega$ as well as fast recovery to minimise switching losses. The fast switching of the MOSFET comes from the fact that the total gate charge is small, hence, charges can be extracted or injected in a small amount of time to reach the desired V_{DS} to turn the MOSFET on or off. The advantage of low losses is two fold: it maximises the efficiency as well as reduce the need of a large heat sink to dissipate wasted power.The low input and output parasitic capacitance of the MOSFET is also helpful in reducing any unwanted ringing in V_{out} when the MOSFET switches on and off.[9]

The chosen MOSFET has a large margin to its absolute maximum ratings, so that in more extreme conditions, such as input over-voltage, the MOSFET will not be destroyed.

4.3.1 MOSFET Gate Drive

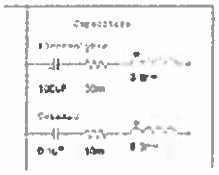
In order to utilise the fast switching capabilities of the MOSFET selected, a gate driver must be used in order to pull and push charges onto the gate as quickly as possible. *Micrel MIC4120* was good for the purpose of driving a low-side gate at a peak current of 6A. The driver has a small delay, which is desirable. There is also a dual driver variant with an inverting and non-inverting output, which will become useful in a half-bridge application.[10]

4.4. Diode

When choosing the diode, metal-semiconductor Schottky junctions instead of a traditional pn junction diode were chosen. This is because the Schottky diode relies on the majority carriers, instead of the minority carriers as in a pn junction, the forward voltage drop is much less at around 0.3V, as opposed to about 0.6V respectively. The Schottky diode also has a faster recovery speed.



(a) Inductor



(b) Capacitor

Fig. 11: PSPICE Models

To benefit from this, a Schottky diode was chosen to meet the electrical requirements as follows: (i) the diode's $I_{max} = 17A > I_{in} + \frac{1}{2}\Delta i_L \approx 1.7A$, as well as the maximum reverse voltage to be bigger than V_{out}, which is the same as the MOSFET.

The *ON Semiconductor NTST30100CTG* Schottky diode with $I_{max} = 15A$ and $V_{R(max)} = 100V$ matched the electrical properties criteria.[11] Its very low forward voltage drop of $V_F = 0.455V$ made this choice especially attractive, as the power loss from the diode can be minimised to increase efficiency.

4.5. PSPICE Component Modelling

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4.5.1 Inductor

For the inductor model, the parasitic capacitance and resistance (ESR) of the inductor will be included, which can be found from the datasheet and calculated from the self-resonance frequency via $f = \frac{1}{2\pi\sqrt{LC}}$.[12] From this the component can be modelled as 3 connected components, as seen in Figure 11a.

4.5.2 Capacitor

Similarly for the capacitor model, the parasitic resistance (ESR) and inductance (ESL) of the inductor were included from the datasheet and calculated from the self-resonance frequency as before, respectively.[13] From this the component can be modelled as 3 components in series, as seen in Figure 11b.[13]

4.5.3 MOSFET and Diode

To model the switch, the PSPICE model included in the library was used. The Schottky diode model used in PSPICE was for *ON Semiconductor MBRS260T3*, which has a similar performance seen previously for *ON Semiconductor NTST30100CTG* albeit a slightly higher forward voltage drop. As a result, the simulation will now have a slightly lower efficiency.

5. Simulation of Converter with Real Components

The real components are then modelled and simulated in PSPICE.



Fig. 12: Schematic of Converter with Real Components

5.1. Efficiency with Varying Load

As expected, the efficiency when using real components is generally lower due to the presence of parasitic resistances, capacitances and inductances.

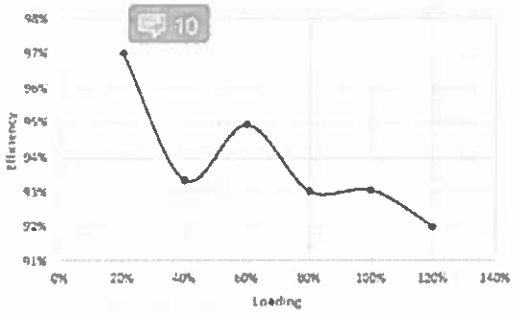


Fig. 13: Graph of Efficiency against Loading

Unlike using ideal components, the efficiency of the converter with real components generally decreases with increasing loading. As the loading increases, the current through the circuit increases to maintain a constant output voltage in continuous conduction. Thus, more power loss is generated by the parasitic resistances of the capacitor and inductor, reducing efficiency.

5.2. Efficiency with Varying Input Voltage

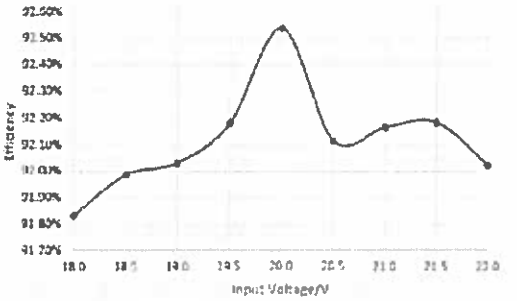


Fig. 14: Graph of Efficiency against input voltage

The efficiency of the converter is maximum at 20V and is lower for other input voltages. As the circuit components were designed for $V_{in} = 20V$, it is expected that the optimal efficiency of the components will be at that operating point.

When the voltage increases beyond this operating point, the input ripple current increases (based on inductor parameters), causing greater power loss across the ESR of inductor, therefore efficiency decreases.

When the voltage decreases beyond this operating point, the peak inductor current decreases, therefore less energy is stored in the inductor and transferred to the capacitor, decreasing the efficiency.

5.3. Converter at Turn on

5.3.1 Overshoot of Output Voltage

The overshoot of output voltage when using real components decreases with increasing input voltage and loading. When the input voltage increases, the input ripple current increases. Since $I_{cap} = I_{L,avg} + \frac{1}{2}\Delta I_L$, the current across the capacitor increases,

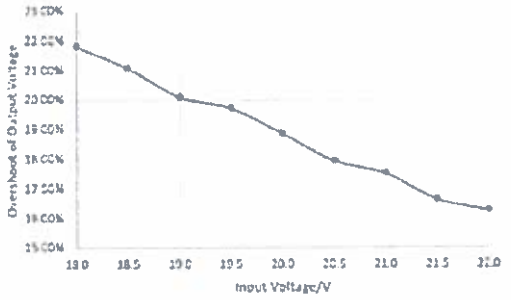


Fig. 15: Graph of Overshoot against input Voltage

In the case of increasing load, the trend is similar to that seen for ideal components in Figure 4. Since the ESR is fixed based on the component parameters, as explained earlier, the decrease in line impedance causes the overshoot to decrease.

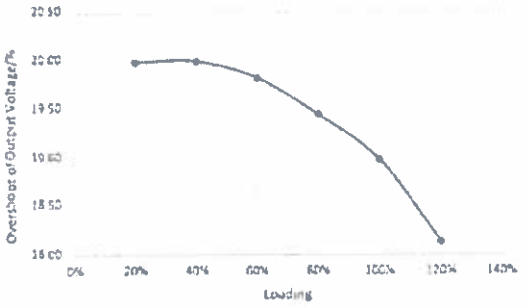


Fig. 16: Graph of Overshoot against Loading

The overshoot when using real components is generally lower than when using ideal components. This is due to the presence of parasitic resistances in the inductor and capacitor that limit the inrush current.

5.3.2 Rise and Settling Time of Output Voltage

The rise and settling time decrease with increasing loading. With increasing input voltage, the rise and settling time remain at approximately 1.40ms and 4.85ms respectively.

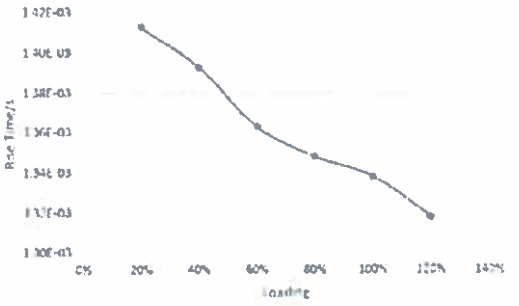


Fig. 17: Graph of Rise Time against Loading

The decreasing trend seen in Figure 16 and 17 follows that of the ideal components. As explained earlier, the decrease in line impedance reduces the rise and settling times.

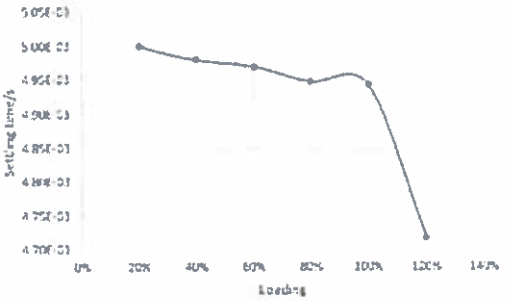


Fig. 18: Graph of Settling Time against Loading

The rise and settling times seen here are larger than that seen for a converter with ideal components. This is due to the parasitic capacitances and inductances in the circuit that cause further delay in the response.

5.3.3 Oscillation Frequency of Output Voltage

The oscillation frequency for the given input range is constant at 245Hz. This is much smaller than the oscillation frequency of the ideal converter. The presence of parasitic resistances damp the response of the LC circuit, thus reducing the oscillation frequency.

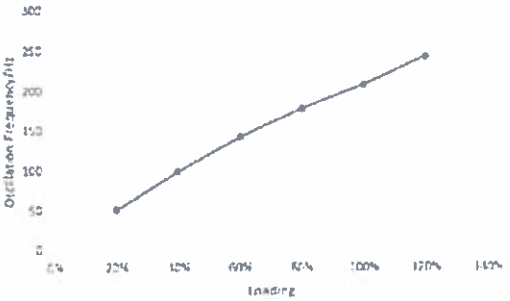


Fig. 19: Graph of Oscillation Frequency against Loading

As for increasing loading, the oscillation frequency increases as seen with the ideal converter, therefore the same explanation applies.

5.4. *Effects of Switching Loads*

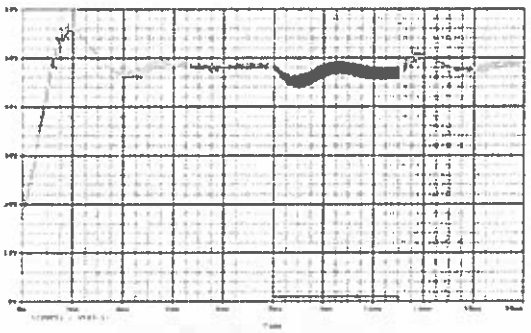


Fig. 20: Simulation of Output Volatge with Switching of Loads from 100% - 50% - 100%

As seen in Figure 20, the area highlighted in black indicates the switching of load from 100% to 50%. The ringing is reduced due to the damping introduced by the parasitic resistances. However, a closed-loop system is still essential in producing a constant output voltage.

5.5. *Effects of Very Low Loads*

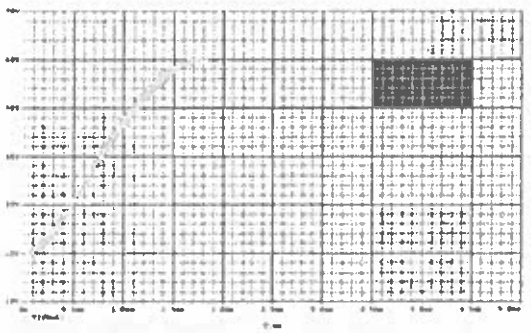


Fig. 21: Simulation of Output Voltage with Very Low Loads

As seen before, the converter has been pushed into discontinuous conduction, thus changing the output voltage. However, the output voltage has decreased due to the presence of the ESR of the capacitor that is in parallel with the load, reducing the effective output resistance seen by the circuit, which reduces the output voltage.

6. Synchronous Rectification

In an attempt to further increase the efficiency from the boost converter from about 92%, the sync FET design is used instead, with a half-bridge to replace the MOSFET-diode switch scheme as seen in section 5.

The overlap of both the low-side and high-side MOSFET being switched on at the same time must be avoided, as this would cause the MOSFET to be destroyed from very high current, through the low resistance path provided, as the output voltage rails are short circuited, which is also known as a shoot-through condition.

This is prevented by inserting a dead-time between the switching off of one MOSFET to the switching on of the other MOSFET. The dead-time delay corresponds to the sum of the MOSFET's switching on time and the switching off time, which is about 100 ns. This delay is normally implemented via a half-bridge driver; however, to ease the speed of simulation, the gate driver was implemented via the use of VPULSE in PSPICE. In order to compensate for the dead-time, the effective duty cycle was increased from the 58% used previously to now 66%.

The simulation circuit schematic can be seen in Figure 22.



Fig. 22: Simulation Circuit Schematic

6.1. *Discontinuous Mode Operation*

An issue was foreseen with the basic sync FET design as shown in Figure 22. In discontinuous conduction mode, the inductor current will reach zero whilst the high-side MOSFET is on, i.e. it has discharged . This mode is occurs when $I_{in} < \frac{1}{2}\Delta I_L$. This means that the voltage across the inductor will reach zero before the high-side MOSFET is turned back off, as the inductor is now discharged. The high-side MOSFET is still turn on, hence a negative current could now flow from the capacitor to the inductor. This is undesired as it does not emulate the diode rectification behaviour, as would have seen in section 5. To achieve this, a current sensing circuit must be implemented in order to switch off the high-side MOSFET when the inductor current reaches zero, before the negative current starts conducting. However, as both switches turn off, the inductor output would now float, which may cause unwanted oscillation with the parasitic capacitances of the MOSFETs and the inductor pair. The resulting oscillation voltages could damage the MOSFETs by exceeding the maximum reverse voltage of the MOSFETs. This could be reduced by using an RC pair between the switching node and ground.[14] As the capacitance of the chosen MOSFET is very small, this has not been implemented in our simulation.

6.2. *Simulation Results*

The synchronous converter was then tested in PSPICE, to see whether efficiency has improved over a range of operating condition. Figure 23 shows the output voltage in time as the Sync FET converter is switched on. From this, one can see that the voltage overshoot previously observed

has been largely damped, by the dead-time between the high-side and low-side MOSFET conduction. This is because the always-on conduction path from the input to the output no longer exist, as the high-side MOSFET turns off regularly. As a result, the inrush current is now limited, as well as the voltage overshoot. This is desired to avoid harmonic resonance in the input supply as well as preventing damage to the output circuit caused from over-voltage.

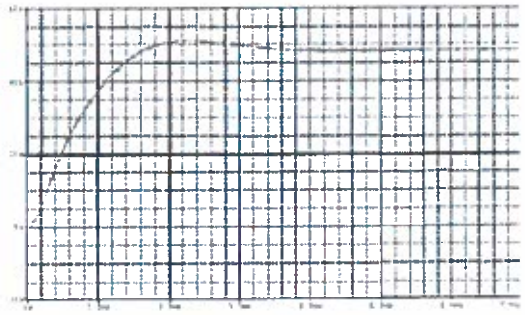


Fig. 23: Simulation Output VOltage

6.2.1 *Varying Load Condition*

The efficiency under various output current will first be examined. The result of this can be seen in figure 24a. Under full load, the efficiency of the sync FET boost converter is 94%, which is an improvement from 92% seen from section 5 without the high-side MOSFET. The efficiency can be seen to increase as load decease, up to around 99%, which can be explained as there are less current passing through the parasitic resistance of the components in the converter, hence there are less losses in the system, so efficiency increases.

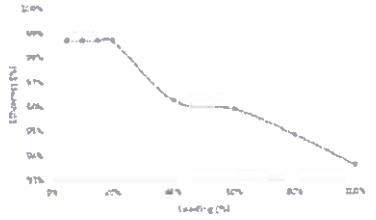
The output voltage of the Sync FET converter decreases as the loading increases as seen in previous simulations. This is because the duty cycle is fixed, hence a constant amount of charge is pumped from the input to the output. Without close-loop compensation or a zener diode at the output, the output voltage continues to raise without the 0.6A output current which the fixed duty cycle was designed for. This can be seen in figure 25a.

The simulation showed that the design worked as expected even in discontinuous conduction mode without the aforementioned caution being taken.

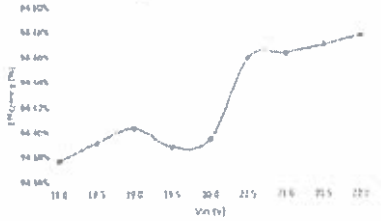
6.2.2 *Varying Input Voltage*

The efficiency of varying input voltage showed less interesting results, despite that a trend found in section 5. Our simulation result showed that the efficiency was almost constant at 94.4% despite different input voltages. This can be seen in figure 24b. There is a very slight slope which shows increase efficiency with increasing voltage. This could be caused by more energy being transfered by the inductor when the input voltage increases.

The output voltage increase when the input voltage increase as expected. This is seen in figure 25b. This is again because the fixed duty cycle in the open loop system

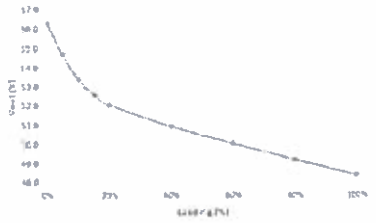


(a) Against Varying Load

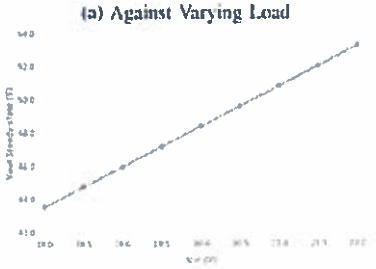


(b) Against Varying Input Voltage

Fig. 24: Graphs of SyncFET Efficiency Results



(a) Against Varying Load



(b) Against Varying Input Voltage

Fig. 25: Graphs of SyncFET Output Voltage Results

does not compensate for the rise in the input voltage, and hence continue to operate the boost converter at the same transfer ratio. From this, it is obvious to see the need of close-loop compensation to regulate the output voltage when operating condition changes. The boost circuit design have now reached a stage where designing the compensator to close the loop can be carried out.

7. Closed Loop System Design

7.1. Static and Dynamic Converter Model

7.1.1 Non Zero Diode Drop

For a non-zero diode drop,

$$V_{out} = V_{in}\left(\frac{1}{1-\delta}\right) - V_{diode}$$

For the diode chosen, $V_{diode} \approx 0.45V$ and can therefore be neglected in the state space averaging model as it is negligible compared to the output voltage. Eventually, the closed loop system will adjust the duty cycle to compensate for this loss.

7.1.2 State Space Averaging Model

Based on the Middlebrook and Ćuk technique,[15] it is possible to model the system whilst ignoring switching frequency ripple. The model can be implemented in Matlab to analyse the open loop characteristics of the system(See Appendix for code).

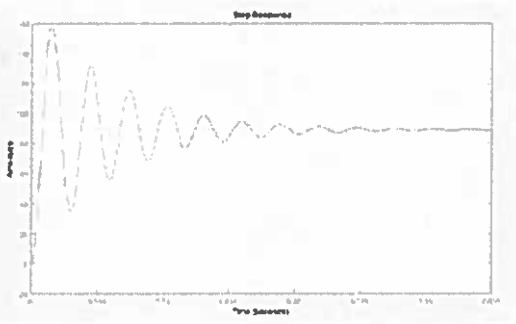


Fig. 26: Step Output of Open Loop System

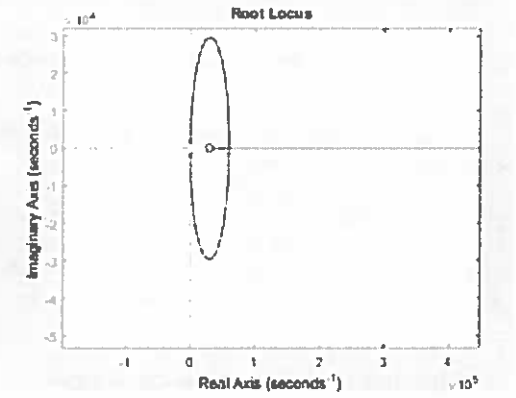


Fig. 27: Root Locus of Open Loop System

The open loop system is stable with poles at $-156 \pm 2120i$. However, the the step response shows large overshoot and bad transient response.

7.2. Compensator Transfer Function

For good regulation of output, the system requires integral action for zero steady state error and good error tracking.[16]

To soft start the converter and limit the inrush current, a resistor in parallel to a MOSFET can be connected to the input stage of the converter. The resistor helps to limit inrush current while the MOSFET will be switched off until steady state voltage is reached. This requires the MOSFET to be controlled a comparator circuit that compares the output voltage with the reference voltage.

To limit the overshoot in output voltage(increase steady state accuracy), the system would require damping which is achieved through a large closed-loop bandwidth.[16] However, this increases the speed of the response and a compromise is required.

7.2.1 Compensator Design

The closed loop system with K=1 is unstable as shown in Figure 26 as the poles have non-negative real parts.

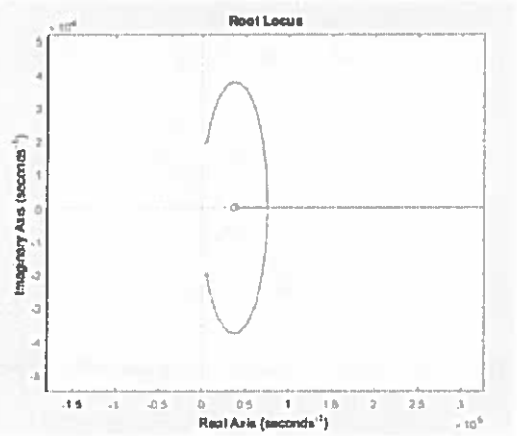


Fig. 28: Root Locus of Closed Loop System with no Compensation

A simple proportional controller with $K_i 1$ would reduce the overshoot and the increase the stability margins but will also increase the rise time, making the system less responsive. A better design solution would be to use a dynamic compensator.

As mentioned before, the system requires an integrator for zero steady state error. It also requires a phase-lead compensator to push the root locus towards the left and improve stability margins.[16] This can be achieved with a proportional-integral controller(PI) in series with a phase-lead controller. The PI controller would help eliminate steady state error while the phase-lead controller would help stabilise the system.The coefficients for the PI controller are chosen by using the PID tuner in Matlab while the coefficients for the lead-lag controller are tuned manually(See Appendix for Code).

The final closed loop design has no poles with positive real part, thus making the system stable. The coefficients selected are a compromise between a fully damped response and response time of the system.

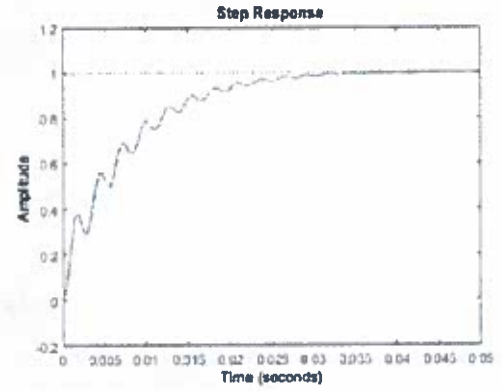


Fig. 29: Step Response of Closed Loop System with Compensation

7.3. *Closed Loop PSpice Model*

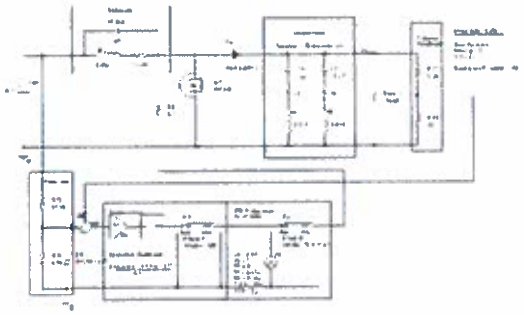


Fig. 30: Schematic of Closed Loop System with Compensation

7.3.1 *PSPICE Simulation*

The PSPICE Simulation schematic can be seen in figure ??.

The above transfer function was then tested with PSPICE simulation. The feedback loop was implemented using a potential divider to take a $\frac{4.7}{48}$ of the output voltage, which is then compared to a 4.7 V reference generated by a zener diode, fed by the input voltage. The difference of the voltage is then multiplied by an integrator with a gain factor of $\frac{48}{4.7}$ to account for the potential divider. The output of this is then fed into a **LAPLACE** block to generate the necessary input into the PWM modulator. The transfer function of the block is derived from the results of section 7.2. The PWM modulator then take this and compare it to the triangular wave generated by **VPULSE** and generate the appropriate duty cycle.

The simulation was then used as a testbed to observe the behaviour of the close-loop system when it is disturbed from the assumptions used to generate the compensator, namely a change in the input voltage, and a change in output current. This will change the duty cycle accordingly, thus moving the operating point, which was assumed constant.

7.3.2 *Simulation Results*

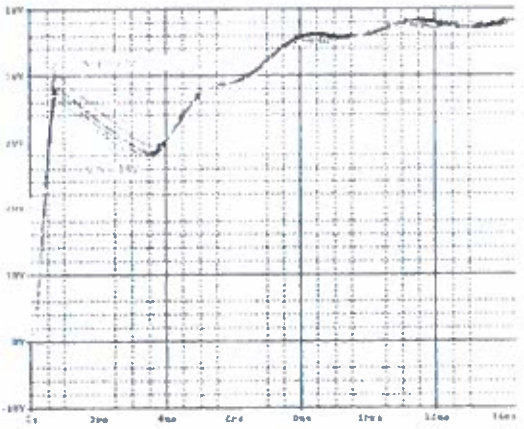


Fig. 31: Close-loop Output Voltage at Varying Input Voltage

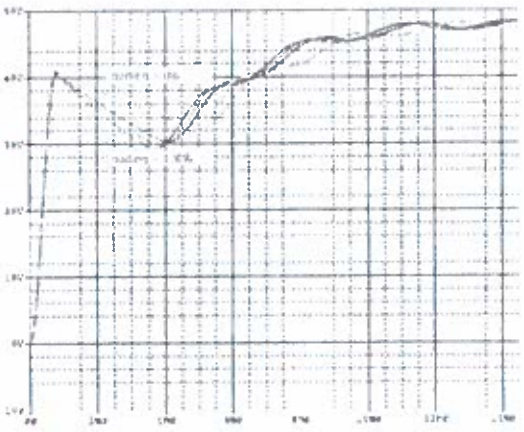


Fig. 32: Close-loop Output Voltage at Varying Loading

The output voltage against time for different operating conditions can be seen in figures ?? and ??.

As seen from the result, there is a slight spike of the output voltage initially at switch on, whilst the compensator is becoming active. Then the output voltage is regulated tightly by the feedback loop to stay around 48V with a ripple of around ± 0.1V. This meets the specification.

7.3.3 *Output Voltage against Varying Input Voltage*

The primary reason for implementing the close loop regulation is to control the output voltage regardless of input voltage. This can be clearly seen in the output voltage against input voltage, where the output voltage stays tightly around 48V.

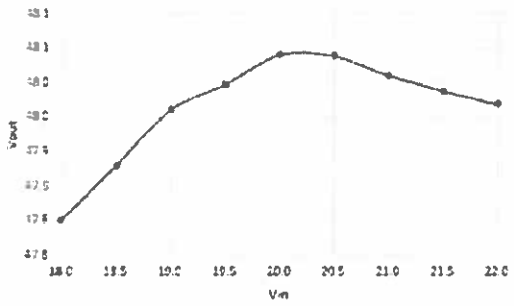


Fig. 33: Graph of Output Voltage against Input Voltage

7.3.4 *Efficiency against Input Voltage*

The efficiency seen with the closed loop model is greater than that of the open loop model because the duty cycle is adjusted such that the output voltage remains at around 48V. Since the system is designed for this output voltage, loss of efficiency from operating at different operating points is reduced. Also, the closed loop system reduces the power loss due to inrush current through the ESR of the capacitor.

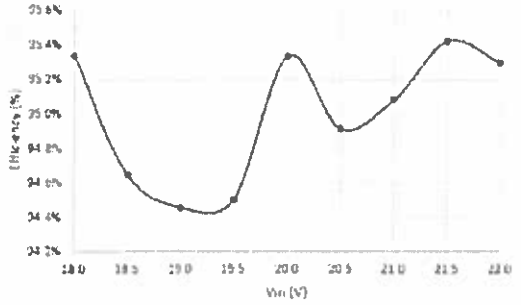


Fig. 34: Graph of Efficiency against Input Voltage

7.3.5 *Output Voltage with Varying Load*

Figure 32 shows no overshoot and the achievement of a steady state output at around 12ms. This shows that the compensator is effective is eliminating the overshoot and steady state error. The response time is also acceptable, thus showing a compromise has been made between the transient response and steady state accuracy.

Moreover, the change in load has little impact on overshoot,rise time and settling time, further showing the compensator is effective in handling different

8. Conclusion

The analysis of the boost converter has given a better insight as to modelling it in real life as will be accomplished in Part 2 of the coursework. Part 1 has provided evidence of the benefits of using a closed loop system and synchronous rectification to improve the efficiency of the system.

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APPENDIX I
APPENDIX

Matlab Code

```
% State-space average model of a Boost Converter
% u=[delta]; x=[i_L; v_C]; y=[v_O]
% Component Parameters
L=470e-6; r_L=100e-3; C=100e-6; r_C=80e-6; R=80;
% Circuit Conditions
V_I=20; Delta=0.53;

% First form models of the two states
A_on = [-r_L/L,  0;  0,  -1/C/(R+r_C)];
B_on = [1/L;  0];
C_on = [0,  R/(R+r_C)];
A_off = [-r_L/L-r_C*R/L/(R+r_C), -R/L/(R+r_C); R/C/(R+r_C), -1/C/(R+r_C)];
B_off = [1/L; 0];
C_off = [r_C*R/(R+r_C), R/(R+r_C)];
% Average the two models
% Operating point model
A = Delta*A_on + (1-Delta)*A_off;
B = Delta*B_on + (1-Delta)*B_off;
C = Delta*C_on + (1-Delta)*C_off;
U=V_I;
X = -inv(A)*B*U;
Y = C*X;
%Small-signal model
E = (A_on-A_off)*X + (B_on-B_off)*U;
F = (C_on-C_off)*X;
boost_ss = ss(A,E,C,F);
% Find unit step-response
T = 0:0.001:0.05;
[y,T,x] = step(boost_ss,T);
% Scale step-response and add to operating point
iL = 0.05*x(:,1)+X(1)*ones(length(T),1);
y = 0.05*y+Y*ones(length(T),1);
% Add time prior to step
T = [0; T+0.15];
iL = [X(1); iL];
y = [Y; y];



z = -1000;
p_plus_i = tf([1,-z],[1, 0]);
piboost = series(p_plus_i, boost_ss);
k=0.0012;
piboost_closed = feedback(k*piboost, 1);
%
Kc=1.25;p=-100;z1=-80;
C_lead = tf([1,-z1],[1, -p]);
piboost = series(C_lead, piboost_closed);
sysA = Kc*piboost;
step(sysA);

% Plot results to same scale as pSpice simulation
% subplot(2,1,1)
% plot([0, 0.15, 0.1501, 0.2], [0.5, 0.5, 0.55, 0.55])
% axis([0.145, 0.195, 0.45, 0.6])
% % subplot(3,1,2)
```

```
plot(T,iL)
axis([0.145, 0.195, 1.5, 2.5])
subplot(2,1,2)
plot(T,y)
axis([0.145, 0.195, 21, 70])
```

Power

GRADEMARK REPORT

FINAL GRADE

/100

GENERAL COMMENTS

Instructor

PAGE 1

Comment 1

Total grade: A+

Converter design: A
Open Loop: A
Component Choices: A
Real Components: A
Synchronous Rectification: A
Closed Loop: A
Additional Marks: A

PAGE 2

PAGE 3

PAGE 4

QM

Good

Comment 2

you should have used the maximum duty cycle value because this capacitance will result in greater ripple at δmax

Comment 3

diode current conducting capability

QM

Correct equation

Additional Comment you used the correct equation but you should have used the maximum δ as at the moment your calculated inductance will result in a current ripple higher than the specs when δ will equal to δmax.

Comment 4

the correct way to present this results is to mention that the maximum voltage blocking capability should be greater than vout,max and current conducting capability should be greater than iin,max

PAGE 5

Comment 5
you should assume an output load smaller or equal to 100%

Comment 6
the lower the input voltage the higher the input current to reach the same power output (48V at 80Ω) hence the greater the losses in the diode and the switch.

PAGE 6

QM

Good

PAGE 7

Comment 7
you can find a 10uF capacitor with 50V or even a 100V voltage blocking capability

Comment 8
low input capacitance of the mosfet implies low power required for switching the mosfet.

Comment 9
Schottky didoes do not experience reverse recovery effects.

PAGE 8

Comment 10
it is quite common to see the efficiency of a dc/dc converter to peak at lower loads than 100% and reach its minimum at 0% load. I would expect the peak to be somewhere between 50-75% load.

QM

Good

PAGE 9

Comment 11
capacitors block the flow of dc current, therefore the current through your cap is just the ripple in the inductor current. when the ripple in that current peaks you should expect to see your peak ripple at your output voltage.

PAGE 10

PAGE 11

PAGE 12

Comment 12
I would avoid adding the resistor there as it will drop your efficiency.

PAGE 13

PAGE 14

PAGE 15

PAGE 16